Algebraic Curves
Chapter 7.4. Quadratic Transformations

The previous procedure produces a new curve with better singularities, but is no longer planar. If we want a newer curve that is planar then we would need more singularities. However these new singularities may be ordinary multiple points, and the old singularities still become better. This motivates the quadratic transformation.

Let $\mathbf{p} = [0:0:1]$, $\mathbf{p}' = [0:1:0]$, and $\mathbf{p}'' = [1:0:0]$; these are the 'fundamental points'. Let $\mathbb{L} = \{Z = 0\}$, $\mathbb{L}' = \{Y = 0\}$, and $\mathbb{L}'' = \{X = 0\}$; these are the 'exceptional lines'. Let $\mathbb{U} = \mathbb{P}^2 \smallsetminus \{XYZ = 0\}$. Define,

$$\mathrm{Q}\colon \mathbb{P}^2 \smallsetminus \{\mathbf{p}, \mathbf{p}', \mathbf{p}''\} \to \mathbb{P}^2, \qquad [x:y:z] \mapsto [yz:xz:xy]$$

Hence Q is a morphism from $\mathbb{P}^2 \smallsetminus \{\mathbf{p}, \mathbf{p}', \mathbf{p}''\}$ to $\mathbb{U} \cup \{\mathbf{p}, \mathbf{p}', \mathbf{p}''\}$. Observe that,

$$\mathrm{Q}^{-1}[\mathbf{p}] = \mathbb{L} \smallsetminus \{\mathbf{p}', \mathbf{p}''\}, \qquad \mathrm{Q}^{-1}[\mathbf{p}'] = \mathbb{L}' \smallsetminus \{\mathbf{p}, \mathbf{p}''\} \qquad \mathrm{Q}^{-1}[\mathbf{p}''] = \mathbb{L}'' \smallsetminus \{\mathbf{p}', \mathbf{p}\}$$

Let $[x:y:z] \in \mathbb{U}$. Hence,

$$\mathrm{Q}[\mathrm{Q}[x:y:z]] = \mathrm{Q}[yz:xz:xy] = [xzxy:yzxy:yzxz] = [x:y:z]$$

Thus Q is an injective endomorphism of $\mathbb{U}$ that is self-inverting, so must be an isomorphism. Hence Q is a birational map of $\mathbb{P}^2$ with itself. It is known as the 'standard quadratic transformation', or the 'standard Cremona transformation'. (Cremona transformations are any birational map of $\mathbb{P}^2$ with itself)

Let $\mathbb{C}$ be an irreducible curve in $\mathbb{P}^2$. Assume $\mathbb{C}$ is not an exceptional line, at which we are done. Then $\mathbb{C} \cap \mathbb{U}$ is open in $\mathbb{C}$ and closed in $\mathbb{U}$. Hence $\mathrm{Q}^{-1}[\mathbb{C} \cap \mathbb{U}] = \mathrm{Q}[\mathbb{C} \cap \mathbb{U}]$ is a closed curve of $\mathbb{U}$. Let $\mathbb{C}'$ be its closure in $\mathbb{P}^2$. Observe that Q restricts on $\mathbb{C}' \smallsetminus \{\mathbf{p}, \mathbf{p}', \mathbf{p}''\}$ to a birational morphism with $\mathbb{C}$, and that applying this same procedure again on $\mathbb{C}'$ produces $\mathbb{C}$ as Q is self-inverting. Algebraically let $\mathrm{F} \in \Bbbk[X, Y, Z]$ be the equation of $\mathbb{C}$ and n its degree. We call $\mathrm{F} \circ \mathrm{Q} = \mathrm{F}[YZ, XZ, XY]$ the 'algebraic transform' of F. Observe that F is of degree $2n$. We prove several properties about this transform.

Claim 1. Let $\mathrm{m}_{\mathbf{p}}[\mathbb{C}] = r$. Then Z^r is the largest power of Z that divides $\mathrm{F} \circ \mathrm{Q}$. Proof. Observe that we may factor out the Z component of F for,

$$\mathrm{F}[X, Y, Z] = \mathrm{F}_r[X, Y]Z^{n-r} + \cdots + \mathrm{F}_n[X, Y]$$

Where F_i is a form of degree i. Hence,

$$\mathrm{F} \circ \mathrm{Q} = \mathrm{F}_r[YZ, XZ](XY)^{n-r} + \cdots = Z^r(\mathrm{F}_r[Y,X](XY)^{n-r} + Z\,\mathrm{F}_{r+1}[Y,X](XY)^{n-r-1} + \cdots)$$

As desired. ■

Let $\mathrm{m}_{\mathbf{p}}[\mathbb{C}] = r$, $\mathrm{m}_{\mathbf{p}'}[\mathbb{C}] = r'$, and $\mathrm{m}_{\mathbf{p}''}[\mathbb{C}] = r''$. Applying the previous result we may factor $\mathrm{F} \circ \mathrm{Q} = Z^r Y^{r'} X^{r''}\,\mathrm{F}'$, where F' is indivisible by Z^r, $X^{r'}$, and $X^{r''}$. The polynomial F' is known as the 'proper transform' of F.

Claim 2. $\deg[\mathrm{F}'] = 2n - r - r' - r''$, the proper transform of F' is F', F' is irreducible, and $\{\mathrm{F}' = 0\} = \mathbb{C}'$. Proof. By computation we have that $\mathrm{F} \circ \mathrm{Q} \circ \mathrm{Q} = (XYZ)^n\,\mathrm{F}$. Thus as F is itself irreducible it follows that F' is irreducible, and shows the proper transform of F' is F'. Lastly $\{\mathrm{F}' = 0\} \supseteq \mathrm{Q}^{-1}[\mathbb{C} \cap \mathbb{U}]$, so necessarily $\{\mathrm{F}' = 0\} = \mathbb{C}'$. ■

Claim 3. $\mathrm{m}_{\mathbf{p}}[\mathrm{F}'] = n - r' - r''$, and similarly for $\mathbf{p}'$ and $\mathbf{p}''$. Proof. Factor,

$$\mathrm{F}' = \sum_{i=0}^{n-r} \mathrm{F}_{r+i}[Y,X] X^{n-r-r''-i} Y^{n-r-r'-i} Z^i$$

Hence the leading form of F' at $\mathbf{p} = [0:0:1]$ is $\mathrm{F}_n[Y,X] X^{-r''} Y^{-r'}$. ■

Claim 4. Define $\mathbb{C}$ to be in 'good position' if no exceptional line is tangent to $\mathbb{C}$ at a fundamental point. Then if $\mathbb{C}$ is in a good position, so too is $\mathbb{C}'$. Proof. Observe that $\mathbb{L}$ is tangent to $\mathbb{C}'$ at $\mathbf{p}'$ *iff*,

$$\begin{aligned}
\langle \mathrm{F}', Z\rangle[\mathbf{p}'] > \mathrm{m}_{\mathbf{p}'}[\mathbb{C}'] &\Leftrightarrow \langle \mathrm{F}_r[Y,X] X^{n-r-r''} Y^{n-r-r'}, Z\rangle[\mathbf{p}'] > n - r - r'' \\
&\Leftrightarrow \langle \mathrm{F}_r[Y,X], Z\rangle[\mathbf{p}'] > 0 \\
&\Leftrightarrow \mathrm{F}_r[1,0] = 0
\end{aligned}$$

However if Y is not tangent to F at $\mathbf{p}$, then $\mathrm{F}_r[1,0] \neq 0$. Similar arguments apply for the other points. ■

Now let $\mathbb{C}$ be in a good position and $\mathbf{p} \in \mathbb{C}$. Define $\mathbb{C}_0 := (\mathbb{C} \cap \mathbb{U}) \cup \mathbf{p}$ a neighbourhood of $\mathbf{p}$ on $\mathbb{C}$. Define $\mathbb{C}_0' := \mathbb{C}' \setminus \{XY = 0\}$ and let $\mathrm{f} : \mathbb{C}_0' \to \mathbb{C}_0$ be the restriction of Q to $\mathbb{C}_0'$ Define $\mathrm{F}_* = \mathrm{F}[X,Y,1]$ and $\mathbb{C}_* = \{\mathrm{F}_* = 0\} \subseteq \mathbb{A}^2$ their dehomogenization. We have their proper transform $(\mathrm{F}_*)' = \mathrm{F}[X, XZ, 1] X^{-r}$, $\mathbb{C}_*' = \{\mathrm{F}_*' = 0\} \subseteq \mathbb{A}^2$ and $\mathrm{f}_* : \mathbb{C}_*' \to \mathbb{C}_*$, $(x,z) \mapsto (x, xz)$ in parallel.

Claim 5. There exists a neighbourhood $\mathbb{W} \subseteq \mathbb{C}_*$ of $(0,0)$ and $\mathrm{f}_*^{-1}[\mathbb{W}] =: \mathbb{W}'$, and isomorphisms $\varphi : \mathbb{W} \to \mathbb{C}_0$, $\varphi' : \mathbb{W}' \to \mathbb{C}_0'$ s.t. $\varphi \circ \mathrm{f}_* = \mathrm{f} \circ \varphi'$. Namely we have the diagram,

$$\begin{tikzcd}
\mathbb{C}_*' \supseteq \mathbb{W}' \arrow[r, "\varphi'"] \arrow[d, "\mathrm{f}_*"] & \mathbb{C}_0' \subseteq \mathbb{C}' \arrow[d, "\mathrm{f}"] \\
\mathbb{C}_* \supseteq \mathbb{W} \arrow[r, "\varphi"] & \mathbb{C}_0 \subseteq \mathbb{C}
\end{tikzcd}$$

Proof. Let $\mathbb{W} = (\mathbb{C}_* \setminus \{XY = 0\}) \cup (0,0)$, $\varphi[x,y] = [x : y : 1]$, and $\varphi'[x,z] = [z : 1 : xz]$. The inverse of φ' is therefore $[x : y : z] \mapsto [z/x, x/y]$. ■

Claim 6. If $\mathbb{C}$ is in good position and $\mathbf{p}_1, \ldots, \mathbf{p}_s$ are the non-fundamental points on $\mathbb{C}' \cap \mathbb{L}$, then $\mathrm{m}_{\mathbf{p}_i}[\mathbb{C}'] \leq \langle \mathbb{C}', \mathbb{L} \rangle [\mathbf{p}_i]$ and $\sum_{i=1}^{s} \langle \mathbb{C}', Z \rangle [\mathbf{p}_i] = r$. Proof. Compute,

$$\sum_i \langle \mathrm{F}', Z \rangle [\mathbf{p}_i] = \sum_i \langle \mathrm{F}_r[Y, X], Z \rangle [\mathbf{p}_i] = r$$

■

We now say that $\mathbb{C}$ is in 'excellent position' if it is in good position and $\mathbb{L}$ intersects $\mathbb{C}$ transversally at n distinct non-fundamental points and $\mathbb{L}'$ and $\mathbb{L}''$ intersect $\mathbb{C}$ transversally at $n - r$ distinct non-fundamental points.

Claim 7. If $\mathbb{C}$ is in excellent position then $\mathbb{C}'$ has multiple points at (a) where on $\mathbb{C}' \cap \mathbb{U}$ correspond to those on $\mathbb{C} \cap \mathbb{U}$ preserving multiplicity and ordinariness; (b) $\mathbf{p}$, $\mathbf{p}'$, and $\mathbf{p}''$ are ordinary multiple points on $\mathbb{C}'$ with multiplicity n, $n - r$, and $n - r$ respectively; (c) there are no non-fundamental points on $\mathbb{C} \cap \mathbb{L}'$ or $\mathbb{C}' \cap \mathbb{L}''$ with, where $\mathbf{p}_1, \ldots, \mathbf{p}_s$ the non-fundamental points on $\mathbb{C}' \cap \mathbb{L}$, $\mathrm{m}_{\mathbf{p}_i}[\mathbb{C}'] \leq \langle \mathbb{C}', \mathbb{L} \rangle [\mathbf{p}_i]$ and $\sum_i \langle \mathbb{C}', \mathbb{L} \rangle = r$. Proof. (a) is as by birational morphism of Q we know that $\mathbb{C} \cap \mathbb{U}$ and $\mathbb{C}' \cap \mathbb{U}$ are isomorphism. Hence their multiplicities and ordinary points coincide. (b) follows from the previous claim applied to $\mathbb{C}'$ as $(\mathbb{C}')' = \mathbb{C}$. (c) follows by applying the previous claim to $\mathbb{C}$. ■

Now define the following polynomial for irreducible projective plane curves $\mathbb{C}$ of degree n with multiple points of multiplicity $r_{\mathbf{p}} = \mathrm{m}_{\mathbf{p}}[\mathbb{C}]$,

$$\mathrm{g}^*[\mathbb{C}] = \frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p}} \frac{r_{\mathbf{p}}(r_{\mathbf{p}} - 1)}{2}$$

Claim 8. If $\mathbb{C}$ is in excellent position, then,

$$\mathrm{g}^*[\mathbb{C}'] = \mathrm{g}^*[\mathbb{C}] - \sum_{i=1}^{s} \frac{r_i(r_i - 1)}{2}$$

Where $r_i = \mathrm{m}_{\mathbf{p}_i}[\mathbb{C}']$ and $\mathbf{p}_1, \ldots, \mathbf{p}_s$ are the non-fundamental points of $\mathbb{C}' \cap \mathbb{L}$. Proof. This follows from Claim 2 and Claim 7. ■

With this we are finally able to prove the main results of the chapter.

Proposition. Let F be an irreducible projective plane curve, $\mathbf{p} \in \{\mathrm{F} = 0\}$. Then there is a projective automorphism T s.t. $\mathrm{F} \circ \mathrm{T}$ is in excellent position and $\mathrm{T}[0 : 0 : 1] = \mathbf{p}$.

Proof.

Denote $\deg[\mathrm{F}] = n$ and $\mathrm{m}_{\mathbf{p}}[\mathrm{F}] = r$. Recall from Question 4.16 that it suffices to find lines $\mathbb{L}$, $\mathbb{L}'$, and $\mathbb{L}''$ s.t.,

$$\mathbb{L}' * \mathbb{C} = r\mathbf{p} + \mathbf{p}'_{r+1} + \cdots + \mathbf{p}'_n$$

$$\mathbb{L}'' * \mathbb{C} = r\mathbf{p} + \mathbf{p}''_{r+1} + \cdots + \mathbf{p}''_n$$
$$\mathbb{L} * \mathbb{C} = \mathbf{p}_1 + \cdots + \mathbf{p}_n$$

Where all these points are distinct. This yields a projective automorphism T with $\mathrm{L} \circ \mathrm{T} = Z$, $\mathrm{L}' \circ \mathrm{T} = Y$, and $\mathrm{L}'' \circ \mathrm{T} = X$. This is possible from Question 5.26.

■

Where T is any projective automorphism, $\mathrm{Q} \circ \mathrm{T}$ is denoted a 'quadratic transformation', and the proper transform of $\mathrm{F} \circ \mathrm{T}$ the 'quadratic transformation' of F. If $\mathrm{F} \circ \mathrm{T}$ is in excellent position and $\mathrm{T}[0:0:1] = \mathbf{p}$, then the quadratic transformation is 'centered at $\mathbf{p}$'. Let $\mathrm{F} := \mathrm{F}_1, \mathrm{F}_2, \ldots, \mathrm{F}_m =: \mathrm{G}$ be a sequence of curves where each F_i is a quadratic transformation of F_{i-1}. In this we say that F is transformed into G (through a finite sequence of quadratic transformations).

Proposition. Any irreducible projective plane curve may be transformed into a curve whose singularities are only ordinary multiple points using a finite sequence of quadratic transformations.

Proof.

We inductively apply quadratic transformations, each time centering at a non-ordinary multiple point. Then from Claim 7 and Claim 8 we see that the resultant curve $\mathbb{C}'$ has one less ordinary multiple point than the original $\mathbb{C}$ in that,

$$\mathrm{g}^*[\mathbb{C}'] = \mathrm{g}^*[\mathbb{C}] - \sum_i \frac{r_i(r_i - 1)}{2} < \mathrm{g}^*[\mathbb{C}]$$

Thus if the original curve $\mathbb{C}$ had N non-ordinary multiple points, then at most $N + \mathrm{g}^*[\mathbb{C}]$ quadratic transformations are necessary to produce the ultimate curve.

■

Question 7.10

Proposition.
$\mathrm{F} = 8X^3Y + 8X^3Z + 4X^2YZ - 10XY^3 - 10XY^2Z - 3Y^3Z$ is in good position, and $\mathrm{F}' = 8Y^2Z + 8Y^3 + 4XY^2 - 10X^2Z - 10X^2Y - 3X^3$.

Proof.

We first show that no exceptional line is tangent to F at an exceptional point. By a direct computation first $\mathrm{F}[\mathbf{p}] = \mathrm{F}[\mathbf{p}'] = \mathrm{F}[\mathbf{p}''] = 0$. We now compute the tangents to F. We move onto their affine charts,

$$\mathrm{F}[X, Y, 1] = 8X^3Y + 8X^3 + 4X^2Y - 10XY^3 - 10XY^2 - 3Y^3$$

$$\mathrm{F}[X, 1, Z] = 8X^3 + 8X^3Z + 4X^2Z - 10X - 10XZ - 3Z$$

$$\mathrm{F}[1, Y, Z] = 8Y + 8Z + 4YZ - 10Y^3 - 10Y^2Z - 3Y^3Z$$

At which it is evident that none of the exceptional lines are tangent in the respective affine charts, as none of the dehomogeneizations of F are divisible by X, Y, or Z. This shows that F is in good position.

We next compute F'. First observe that $n = \deg \mathrm{F} = 4$ and from the affine charts that $r = \mathrm{m}_{\mathbf{p}}[\mathrm{F}] = 3$, $r' = \mathrm{m}_{\mathbf{p}'}[\mathrm{F}] = 1$, and $r'' = \mathrm{m}_{\mathbf{p}''}[\mathrm{F}] = 1$. We factor,

$$\mathrm{F} = (8X^3 + 4X^2Y - 10XY^2 - 3Y^3)Z + (8X^3Y - 10XY^3)$$

So that,

$$\begin{aligned}\mathrm{F}' &= (8Y^3 + 4Y^2X - 10YX^2 - 3X^3) + (8Y^3X - 10YX^3)X^{-1}Y^{-1}Z \\ &= 8Y^2Z - 10X^2Z + 8Y^3 + 4Y^2X - 10YX^2 - 3X^3\end{aligned}$$

As desired.

Let us verify that they have the same singularities. For this we compute,

$$\nabla \mathrm{F} = \begin{bmatrix} 24X^2Y + 24X^2Z + 8XYZ - 10Y^3 - 10Y^2Z \\ 8X^3 + 4X^2Z - 30XY^2 - 20XYZ - 9Y^2Z \\ 8X^3 + 4X^2Y - 10XY^2 - 3Y^3 \end{bmatrix},$$

$$\nabla \mathrm{F}' = \begin{bmatrix} 4Y^2 - 20XZ - 20XY - 9X^2 \\ 16XY + 24Y^2 + 8XY - 10X^2 \\ 8Y^2 - 10X^2 \end{bmatrix}$$

And,

$$\nabla \mathrm{F}[0:0:1] = (0,0,0), \qquad \nabla \mathrm{F}[0:1:0] = (-10,0,-3), \qquad \nabla \mathrm{F}[1:0:0] = (0,8,8)$$

$$\nabla \mathrm{F}'[0:0:1] = (0,0,0), \qquad \nabla \mathrm{F}'[0:1:0] = (4,24,8), \qquad \nabla \mathrm{F}'[1:0:0] = (-9,-10,-10)$$

Namely F and F' indeed share the same singularity at $\mathbf{p} = [0:0:1]$. In particular this is a multiple point for both F and F'.

Question 7.11

Observe presently, letting the curve be $\mathrm{F} = Y^2Z - X^3$,

$$\mathrm{F}[X, Y, 1] = Y^2 - X^3, \qquad \mathrm{F}[X, 1, Z] = Z - X^3, \qquad \mathrm{F}[1, Y, Z] = Y^2Z - 1$$

And F intersects $\mathbb{L} = \{Z = 0\}$ at $[0 : 1 : 0]$ with multiplicity one, and nowhere else. It moreover intersects $\mathbb{L}' = \{Y = 0\}$ at $[0 : 0 : 1]$ with multiplicity two, and $\mathbb{L}'' = \{X = 0\}$ at $[0 : 1 : 0]$ and $[0 : 0 : 1]$ at multiplicities one and two respectively.

To be in excellent position we first need $\{\mathrm{F} = 0\}$ to intersect $\{Z = 0\}$ at three non-fundamental points with multiplicity one. Consider the generic projective automorphism,

$$\mathrm{T}[X : Y : Z] = [a_1X + a_2Y + a_3Z : b_1X + b_2Y + b_3Z : c_1X + c_2Y + c_3Z]$$

We require $\mathrm{T}[0 : 0 : 1] = [0 : 0 : 1]$, which constrains it to be,

$$\mathrm{T}[X : Y : Z] = [a_1X + a_2Y : b_1X + b_2Y : c_1X + c_2Y + Z]$$

At which

$$\mathrm{F} \circ \mathrm{T} = (b_1X + b_2Y)^2(c_1X + c_2Y + Z) - (a_1X + a_2Y)^3$$

And,

$$\mathrm{F} \circ \mathrm{T}[1, Y, Z] = (b_1 + b_2Y)^2(c_1 + c_2Y + Z) - (a_1 + a_2Y)^3$$

Additionally,

$$\frac{\partial}{\partial Y}[\mathrm{F} \circ \mathrm{T}] = (b_1 + b_2Y)(2b_2c_1 + 2b_2c_2Y + 2b_2Z + c_2b_1 + c_2b_2Y) + 3a_2(a_1 + a_2Y)^2$$

$$\frac{\partial}{\partial Z}[\mathrm{F} \circ \mathrm{T}] = (b_1 + b_2Y)^2$$

Now so long as $b_1 + b_2Y \neq 0$ and out point $[1 : Y : 0]$ vanishes the first polynomial we are guarenteed that it is simple. Let us choose for simplicity $[1 : -1 : 0]$, so that $1 = b_1 \neq b_2 = -1$ and we are to solve $4(c_1 - c_2) - (a_1 - a_2)^3 = 0$. Let us choose $a_1 = 4, a_2 = 2$, so that choosing $c_1 = 4$ and $c_2 = 2$ are solutions. Thus our transformed curve is,

$$\mathrm{F} \circ \mathrm{T} = (X - Y)^2(4X + 2Y + Z) - (4X + 2Y)^3$$

With,

$$\nabla[\mathrm{F} \circ \mathrm{T}] = \begin{bmatrix} 2(X - Y)(4X + 2Y + Z) + 8(X - Y)^2 - 12(4X + 2Y)^2 \\ -2(X - Y)(4X + 2Y + Z) + 2(X - Y)^2 - 6(4X + 2Y)^2 \\ (X - Y)^2 \end{bmatrix}$$

Evidently the resultant curve is still in good position as none of X, Y, nor Z divides it. It intersects with $\{Z = 0\}$ when,

$$\begin{aligned}
(X - Y)^2(4X + 2Y) - (4X + 2Y)^3 = 0 &\Leftrightarrow (4X + 2Y)((X - Y)^2 - (4X + 2Y)^2) = 0 \\
&\Leftrightarrow (4X + 2Y)(X^2 - 2XY + Y^2 - 16X^2 - 16XY - 4Y^2) = 0 \\
&\Leftrightarrow (4X + 2Y)(-15X^2 - 18XY - 3Y^2) = 0 \\
&\Leftrightarrow -6(2X + Y)(X + Y)(5X + Y) = 0
\end{aligned}$$

So intersect at the points $[1:2:0]$, $[1:-1:0]$, and $[1:-5:0]$. Each of these points do not vanish either the first or second coordinate of $\nabla[\mathrm{F}\circ\mathrm{T}]$, so are simple and are also do share Z as a tangent. Moreover observe that $\mathrm{F}\circ\mathrm{T}$ still vanishes at $\mathbf{p}=[0:0:1]$ with this being a point of multiplicity two. With $\{Y=0\}$,

$$\begin{aligned} X^2(4X+Z)-(4X)^3=0 &\Leftrightarrow 4X^3+X^2Z-64X^3=0 \\ &\Leftrightarrow X^2(-60X+Z)=0 \end{aligned}$$

So that it intersects non-fundamentally at $[1:0:60]$, and this point does not vanish $\nabla[\mathrm{F}\circ\mathrm{T}]$ so is simple. With $\{X=0\}$,

$$\begin{aligned} Y^2(2Y+Z)-(2Y)^3=0 &\Leftrightarrow 2Y^3+Y^2Z-8Y^3=0 \\ &\Leftrightarrow Y^2(6Y-Z)=0 \end{aligned}$$

So that it intersects non-fundamentally at $[0:1:6]$, and this point does not vanish $\nabla[\mathrm{F}\circ\mathrm{T}]$ so is simple. Thus this transformation does indeed bring F into excellent position.

We lastly compute the quadratic transformation. We expand,

$$\begin{aligned} \mathrm{F}\circ\mathrm{T}\circ\mathrm{Q}[X:Y:Z] &= \mathrm{F}\circ\mathrm{T}[YZ:XZ:XY] \\ &= [(X-Y)^2(4X+2Y+Z)-(4X+2Y)^3]\circ[YZ:XZ:XY] \\ &= (YZ-XZ)^2(4YZ+2XZ+XY)-(4YZ+2XZ)^3 \\ &= Z^2(X-Y)^2(4YZ+2XZ+XY)-Z^3(4Y+2X)^3 \end{aligned}$$

And so,

$$(\mathrm{F}\circ\mathrm{T})'=(X-Y)^2(4YZ+2XZ+XY)-8Z(2Y+X)^3$$

Question 7.12

We find a quadratic transformation of $\mathrm{F} = Y^2Z^2 - X^4 - Y^4$ s.t. it only has ordinary multiple points. Let us first understand the multiple points of F. We compute,

$$\nabla \mathrm{F} = (-4X^3, 2YZ^2 - 4Y^3, 2Y^2Z)$$

This tells us that it is necessary that $X = 0$. To vanish the third coordinate either $Y = 0$ or $Z = 0$. In the first case we obtain $[0:0:1] = \mathbf{p}$. In the second case $Y = 0$ which is not a point on $\mathbb{P}^2$. We thus conclude that the only multiple point of F is $[0:0:1]$, with a multiplicity of two. To compute our transform we utilize the generic projective automorphism T about $\mathbf{p}$,

$$\begin{aligned}\mathrm{F} \circ \mathrm{T} &= [Y^2Z^2 - X^4 - Y^4] \circ [a_1X + a_2Y : b_1X + b_2Y : c_1X + c_2Y + Z] \\ &= (b_1X + b_2Y)^2(c_1X + c_2Y + Z)^2 - (a_1X + a_2Y)^4 - (b_1X + b_2Y)^4\end{aligned}$$

Thereby on $\{Z = 0\}$ and dehomogenizing in X,

$$\mathrm{F} \circ \mathrm{T}[1, Y, 0] = (b_1 + b_2Y)^2(c_1 + c_2Y)^2 - (a_1 + a_2Y)^4 - (b_1 + b_2Y)^4$$

A quick computation shows that the choices $a_1 = a_2 = 1, b_1 = 1, b_2 = -1, c_1 = 1, c_2 = -5$ yields four distinct roots. Thus our curve is,

$$\mathrm{F} \circ \mathrm{T} = (X - Y)^2(X - 5Y + Z)^2 - (X + Y)^4 - (X - Y)^4$$

Moreover,

$$\mathrm{F} \circ \mathrm{T}[0, 1, Z] = (-5 + Z)^2 - 2, \qquad \mathrm{F} \circ \mathrm{T}[X, 0, 1] = X^2((X + 1)^2 - 2X^2)$$

Which intersect $\{X = 0\}$ and $\{Y = 0\}$ transversally at two distinct non-fundamental points respectfully. Hence this transformation results in $\mathrm{F} \circ \mathrm{T}$ being in excellent position. Now we compute,

$$\begin{aligned}\mathrm{F} \circ \mathrm{T} \circ \mathrm{Q} &= \mathrm{F} \circ \mathrm{T}[YZ : XZ : XY] \\ &= \mathrm{F}[YZ + XZ : YZ - XZ : YZ - 5XZ + XY] \\ &= (YZ - XZ)^2(YZ - 5XZ + XY)^2 - (YZ + XZ)^4 - (YZ - XZ)^4 \\ &= Z^2(Y - X)^2(YZ - 5XZ + XY)^2 - Z^4(Y + X)^4 + Z^4(Y - X)^4 \\ &= Z^2((Y - X)^2(YZ - 5XZ + XY)^2 - Z^2((Y + X)^4 + (Y - X)^4))\end{aligned}$$

So that,

$$\mathrm{F}' = (Y - X)^2(YZ - 5XZ + XY)^2 - Z^2((Y + X)^4 + (Y - X)^4)$$

Given that F had only one multiple point at $\mathbf{p}$, F' has no multiple points on $\{XYZ = 0\}$. Its multiple points are only at $\mathbf{p}, \mathbf{p}'$ and $\mathbf{p}''$, and these are all ordinary, or on $\{Z = 0\}$, whereby,

$$\mathrm{F}'[X : 1 : 0] = (1 - X)^2X^2$$

So the only other candidate is $[1:1:0]$. Namely we know that it is a point of multiplicity two from the above. More importantly we see that it is in fact an ordinary double point as,

$$\mathrm{F}'[1:Y:Z] = (Y-1)^2Y^2 - Z^2((Y+1)^4 + (Y-1)^4)$$

$$\Downarrow$$

$$\mathrm{F}'[1:U=Y-1:Z] = U^2(U+1)^2 - Z^2((U+2)^4 + U^4)$$

The tangent cone is therefore $U^2 - 16Z^2 = (U-4Z)(U+4Z)$, so there are distinct tangents as desired. Hence all multiple points of F' are indeed ordinary.

We now find a quadratic transformation of $\mathrm{F} = Y^4 + Z^4 - 2X^2(Y-Z)^2$. First consider that its multiple points are at,

$$\nabla \mathrm{F} = (4X(Y-Z)^2, 4Y^3 - 4X^2(Y-Z), 4Z^3 + 4X^2(Y-Z))$$

To vanish the first coordinate either $X = 0$ or $Y = Z$. In the first case it is necessary that $Y = Z = 0$, which is not on $\mathbb{P}^2$. Thus $Y = Z$, so that then $Y = Z = 0$ but $X \neq 0$. Hence our multiple point is $[1:0:0]$. First by a transform it suffices to consider $\mathrm{F} = X^4 + Y^4 - 2Z^2(Y-X)^2$ so that our multiple point is $[0:0:1] = \mathbf{p}$. Now using the generic projective automorphism about $\mathbf{p}$,

$$\mathrm{F} \circ \mathrm{T} = (a_1X + a_2Y)^4 + (b_1X + b_2Y)^4 - 2(c_1X + c_2Y + Z)^2((b_2 - a_2)Y - (b_1 - a_1)X)^2$$

Hence on $\{Z = 0\}$ and dehomogenizing w.r.t Y,

$$0 = (a_1X + a_2)^4 + (b_1X + b_2)^4 - 2(c_1X + c_2)^2((b_2 - a_2) - (b_1 - a_1)X)^2$$

And the choice $a_1 = a_2 = -1$ and $b_1 = b_2 = c_1 = c_2 = 1$ is sufficient to produce unique roots. Hence our curve is,

$$\mathrm{F} \circ \mathrm{T} = 2(X+Y)^4 - 8(X+Y+Z)^2(Y-X)^2$$

Moreover,

$$\mathrm{F} \circ \mathrm{T}[0:Y:1] = Y^2(2Y^2 - 8(Y+1)^2), \qquad \mathrm{F} \circ \mathrm{T}[1:0:Z] = 2 - 8(1+Z)^2$$

So will produce two distinct non-fundamental intersections with $\{X = 0\}$ and $\{Y = 0\}$ respectively. Thereby this transformation places $\mathrm{T} \circ \mathrm{T}$ into excellent position. Hence we may compute,

$$\begin{aligned}
\mathrm{F} \circ \mathrm{T} \circ \mathrm{Q} &= \mathrm{F} \circ \mathrm{T}[YZ : XZ : XY] \\
&= \mathrm{F}[-YZ - XZ : YZ + XZ : YZ + XZ + XY] \\
&= 2(YZ + XZ)^4 - 8(YZ + XZ + XY)^2(XZ - YZ)^2 \\
&= Z^2(2Z^2(Y+X)^4 - 8(YZ + XZ + XY)^2(Y-X)^2)
\end{aligned}$$

And so our proper curve is $\mathrm{F}' = 2Z^2(Y+X)^4 - 8(YZ+XZ+XY)^2(Y-X)^2$. Given that the only multiple point of previous is $[0:0:1] = \mathbf{p}$, the multiple points of F' are $\mathbf{p}, \mathbf{p}', \mathbf{p}''$, and potentially that of,

$$0 = \mathrm{F}'[X:1:0] = -8X^2(1-X)^2$$

Namely $[1:1:0]$ as well. Let us compute its tangents. Namely,

$$\mathrm{F}'[1:Y:Z] = 2Z^2(Y+1)^4 - 8(YZ+Z+Y)^2(Y-1)^2$$

$$\Downarrow$$

$$\mathrm{F}'[1:U=Y-1:Z] = 2Z^2(U+2)^4 - 8(UZ+2Z+U+1)^2U^2$$

So that the tangent cone is $32Z^2 - 8U^2 = 8(2Z-U)(2Z+U)$. The tangent cone splits perfectly, so $[1:1:0]$ is ordinary. Therefore all multiple points on F' are indeed ordinary.

Question 7.13

Part A

Proposition.
In the lemma, we may choose T s.t. given a finite set $S \subseteq \{\mathrm{F} = 0\}$ with $\mathbf{p} \notin S$ we have $\mathrm{T}^{-1}[S] \cap \{XYZ = 0\} = \varnothing$, Thus there is a neighbourhood of S on $\{\mathrm{F} = 0\}$ that is isomorphic to an open set on $(\mathrm{F} \circ \mathrm{T})'$.

Proof.

A generic projective automorphism of $\mathbb{P}^2$ about $\mathbf{p}$ is of the form $\mathrm{T}[X, Y, Z] = [a_1X + a_2Y : b_1X + b_2Y : c_1X + c_2Y + Z]$. By taking the inverse it suffices to prove that there exists coefficients s.t., writing $S = \{(X_i, Y_i, Z_i)\}_i$, $\mathrm{T}[X_i, Y_i, Z_i]$ is not zero in any coordinate for all i. We claim that for a_1 sufficiently large, $a_1X_i + a_2Y_i$ is not zero. Assume for contradiction the contrary. That is, for any a_1, there is (X_i, Y_i, Z_i) s.t. $a_1X_i + a_2Y_i = 0$, so that $X_i = a_2/a_1Y_i$. This is true for infinitely many a_1, which as a_2 is constant implies that there must be infinitely many (X_i, Y_i, Z_i). This contradicts the finiteness of S. This argument works for the other pairs of coefficients as well. This shows existence of our requisite T.

Given that T now maps points not on $\{XYZ = 0\}$, i.e. $\mathbb{U}$, onto S, we see that $\mathrm{F} \circ \mathrm{T}$ vanishes at these points on $\mathbb{U}$. Recall that being still in excellent position it follows that, where $\{(\mathrm{F} \circ \mathrm{T})' = 0\} = \mathbb{C}'$, that $\mathbb{C}' \cap \mathbb{U}$ is isomorphic to $\mathbb{C} \cap \mathbb{U}$ where $\mathbb{C} = \{\mathrm{F} = 0\}$. Hence neighbourhoods of $S \subseteq \{\mathrm{F} = 0\}$ are isomorphic to neighbourhoods of $\mathrm{T}^{-1}[S] \subseteq \mathbb{C} \cap \mathbb{U}$ (by projective automorphism) are isomorphic to neighbourhoods of $\mathbb{C}' \cap \mathbb{U}$.

■

Part B

Proposition.
If S is a finite set of simple points on a plane curve F, then there is a curve F' with only ordinary multiple points and a neighbourhood $U \subseteq \{\mathrm{F} = 0\}$ of S and open set $U' \subseteq \{\mathrm{F}' = 0\}$ of only simple points s.t. $U \cong U'$.

Proof.

By the previous result there is a neighbourhood U of S on $\{\mathrm{F} = 0\}$ that is isomorphic to an open U' set of $(\mathrm{F} \circ \mathrm{T})'$. Namely through a finite sequence of quadratic transformations the resultant curve $\mathbb{C}'$ may have only ordinary multiple points. Now the quadratic transformation is an isomorphism of $\mathbb{U}$ with itself, and U' lies on $\mathbb{U}$, so that the successive quadratic transformations yield a neighbourhood $U^{(n)}$ that is isomorphic to $U' \cong U$ and from the curve $\mathbb{C}'$. It therefore suffices to prove that $U^{(n)}$ comprises only of simple points. Namely as we follow a sequence of isomorphisms to construct $U^{(n)}$, it follows that it suffices that U be a neighbourhood of simple points. This follows as the nonsingular subset of an algebraic curve is open, so we may choose a subset U which contain S. Therefore through our

transformations $U^{(n)}$ is completely simple as desired.

$\blacksquare$

Question 7.14

Part A

We first investigate the degree and effect to g^* of a quadratic transformation at an ordinary multiple point. Note that our quadratic transformation is centered at $\mathbf{p}$, so $\mathrm{F} \circ \mathrm{T}$ is in excellent position. Being a projective automorphism $\mathrm{F} \circ \mathrm{T}$ has the same degree has F. Therefore w.l.o.g. let T be s.t. $\mathbf{p}', \mathbf{p}'' \notin \mathbb{C}$ and $\mathrm{p} = [0:0:1]$. Recall,

$$\deg \mathrm{F}' = 2n - r - r' - r'', \qquad \mathrm{g}^*[\mathbb{C}'] = \mathrm{g}^*[\mathbb{C}] - \sum_i \frac{r_i(r_i - 1)}{2}$$

Where $r = \mathrm{m}_{\mathbf{p}}[\mathrm{F}]$, $r' = \mathrm{m}_{\mathbf{p}'}[\mathrm{F}]$, and $r'' = \mathrm{m}_{\mathbf{p}''}[\mathrm{F}]$ and $r_i = \mathrm{m}_{\mathbf{p}_i}[\mathrm{F}]$ of the non-fundamental points $\mathbf{p}_i$. Hence being centered at $\mathbf{p}$,

$$\deg \mathrm{F}' = 2n - r$$

We consider the distinct tangents at $\mathbf{p}$. Homogeneizing in Z we may write,

$$\mathrm{F} = \prod_{i=1}^{r} \ell_i + \cdots$$

Where ℓ_i are the distinct tangents of F at $\mathbf{p}$. In the chart $X := U, Y := UV$,

$$\prod_{i=1}^{r} \ell_i[X, Y] = \prod_{i=1}^{r} \ell_i[U, UV] = U^r \prod_{i=1}^{r} \ell_i[1, V]$$

So that the transform is $\prod_{i=1}^{r} \ell_i[1, V]$. On the exceptional divisor $U = 0$ we still have $\prod_{i=1}^{r} \ell_i[1, V]$. Each being distinct tangents means that this vanishes at r distinct points $\{(0, v_i)\}_{i=1}^{r}$. The Cremona Map applied after this transformation is $\mathrm{Q}[U, V] = [UV : U : U^2V] = U[V : 1 : UV]$. Hence,

$$\mathrm{Q}[0, v_i] = [v_i : 1 : 0]$$

Each is on $\mathbb{L}$. Thus there are r non-fundamental points on $\mathbb{C}' \cap \mathbb{L}$. Our condition $\sum_i \langle \mathbb{C}, \mathbb{L} \rangle [\mathbf{p}_i] = r$ therefore forces all these non-fundamental points to be simple, i.e. $r_i = 1$ for all i. Therefore,

$$\mathrm{g}^*[\mathbb{C}] - \sum_i \frac{r_i(r_i - 1)}{2} = \mathrm{g}^*[\mathbb{C}]$$

That is to say, $\mathrm{g}^*[\mathbb{C}']$ is unchanged from $\mathrm{g}^*[\mathbb{C}]$.

In the case of a simple point $r = 1$ for an ordinary multiple point. Therefore,

$$\deg \mathrm{F}' = 2n - 1, \qquad \mathrm{g}^*[\mathbb{C}'] = \mathrm{g}^*[\mathbb{C}]$$

In the case of a point not in the zero locus $r = 0$ for an ordinary multiple point. Therefore,

$$\deg \mathrm{F}' = 2n, \qquad \mathrm{g}^*[\mathbb{C}'] = \mathrm{g}^*[\mathbb{C}]$$

Part B

Proposition.
For Question 7.13.B, the curve F' may be of arbitrarily large degree.

Proof.

Recall that in Question 7.13.B our curve F' is such that it has only ordinary multiple points with a neighbourhood U' isomorphic to the neighbourhood U of S from F. In particular our neighbourhood lies on $\mathbb{U}$, so is birationally mapped by quadratic transformations. Hence to get to arbitrary large degrees we simply repeatedly quadratic transform about one of its ordinary multiple points. The previous part guarentees that its degree increases to $2n - r > n$ (r equals n for an n-fold tangent) for the resultant curve, which still has only ordinary multiple points and a neighbourhood isomorphic to U.

■

Question 7.15

Proposition.
Let $\mathrm{F} = \mathrm{F}_1, \ldots, \mathrm{F}_m$ be a sequence of quadratic transformations of F so that F_m has only multiple ordinary points. Let $\mathbf{p}_{i,1}, \mathbf{p}_{i,2}, \ldots$ be the newly introduced points on each F_i from the transformation. Let $n = \deg \mathrm{F}$. Then,

$$(n-1)(n-2) \geq \sum_{\mathbf{p}} \mathrm{m}_{\mathbf{p}}[\mathrm{F}](\mathrm{m}_{\mathbf{p}}[\mathrm{F}]-1) + \sum_{i>1,j} \mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i](\mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i]-1)$$

Proof.

We first prove that the genus inductively expands in consecutive quadratic transformations how we expect. We prove this by induction on m. In the base-case $m = 1$ this is just the definition,

$$\mathrm{g}^*[\mathbb{C}] = \frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p}} \frac{\mathrm{m}_{\mathbf{p}}[\mathrm{F}](\mathrm{m}_{\mathbf{p}}[\mathrm{F}]-1)}{2}$$

Now most generally we have,

$$\mathrm{g}^*[\mathbb{C}_m] = \mathrm{g}^*[\mathbb{C}_{m-1}] - \sum_{j} \frac{\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m](\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m]-1)}{2}$$

So by using the inductive hypothesis on $\mathrm{g}^*[\mathbb{C}_{m-1}]$,

$$\begin{aligned}
&\mathrm{g}^*[\mathbb{C}_{m-1}] - \sum_{j} \frac{\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m](\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m]-1)}{2} \\
&\qquad = \frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p}} \frac{\mathrm{m}_{\mathbf{p}}[\mathrm{F}](\mathrm{m}_{\mathbf{p}}[\mathrm{F}]-1)}{2} \\
&\qquad - \sum_{i=1}^{m-1} \sum_{j=1}^{s_i} \frac{\mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i](\mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i]-1)}{2} - \sum_{j=1}^{s_m} \frac{\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m](\mathrm{m}_{\mathbf{p}_{m,j}}[\mathrm{F}_m]-1)}{2} \\
&\qquad = \frac{(n-1)(n-2)}{2} - \sum_{\mathbf{p}} \frac{\mathrm{m}_{\mathbf{p}}[\mathrm{F}](\mathrm{m}_{\mathbf{p}}[\mathrm{F}]-1)}{2} - \sum_{i=1}^{m} \sum_{j=1}^{s_i} \frac{\mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i](\mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i]-1)}{2}
\end{aligned}$$

Thus proving our lemma. We now recall that the genus $\mathrm{g}^*[\mathbb{C}_m]$ is nonnegative. Thus multiplying out by two and rearranging provides our proposition.

■

Question 7.16

Proposition.
Let F, G be two plane curves with no common components and no multiple components. Then there is a sequence of quadratic transformations $\mathrm{F} = \mathrm{F}_1, \ldots, \mathrm{F}_s = \mathrm{F}', \mathrm{G} = \mathrm{G}_1, \ldots, \mathrm{G}_t = \mathrm{G}'$ s.t. F' and G' have only ordinary multiple points and no tangents in common at points of intersection. Moreover,

$$\deg \mathrm{F} \deg \mathrm{G} = \sum_{\mathbf{p}} \mathrm{m}_{\mathbf{p}}[\mathrm{F}]\, \mathrm{m}_{\mathbf{p}}[\mathrm{G}] + \sum_{i>1} \sum_{j} \mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{F}_i]\, \mathrm{m}_{\mathbf{p}_{i,j}}[\mathrm{G}_i]$$

Where $\mathbf{p}_{i,j}$ are the neighbouring singularities of $\mathrm{F}\,\mathrm{G}$.

Proof.

We apply the resolution of singularities to $\mathrm{F}\,\mathrm{G}$, thereby producing through a sequence of quadratic transformations $\mathrm{Q}_1, \ldots, \mathrm{Q}_k \colon \mathbb{P}^2 \to \mathbb{P}^2$ centered at singularities, a sequence of curves that ultimately result in a curve $(\mathrm{F}\,\mathrm{G})'$ with only ordinary multiple points. We consider applying $\mathrm{Q}_1, \ldots, \mathrm{Q}_K$ to F and G individually. In particular we claim that the resultant curves F' and G' also has only ordinary multiple points and no tangents in common at points of intersection. For this we make an observation. Observe that generally the sequence of quadratic transformations is the Cremona map Q applied after a projective automorphism T that puts the curve into excellent position. We then take the irreducible component on $\mathbb{U}$. Thus most generally define the map,

$$\mathrm{H}_k \colon \Bbbk[\mathbb{P}^2] \to \Bbbk[\mathbb{P}^2], \qquad \mathrm{F} \mapsto \mathrm{F} \circ \mathrm{Q} \circ \mathrm{T}_k \,/ X^{n_{k,1}} Y^{n_{k,2}} Z^{n_{k,3}}$$

And so we may write,

$$(\mathrm{F} \circ \mathrm{G})' = (\mathrm{F}\,\mathrm{G}) \circ \mathrm{H}_1 \circ \cdots \circ \mathrm{H}_K = (\mathrm{F} \circ \mathrm{H}_1 \circ \cdots \circ \mathrm{H}_K)(\mathrm{G} \circ \mathrm{H}_1 \circ \cdots \circ \mathrm{H}_K)$$

Observe that the individual terms of the product are respectively quadratic transformations of F and Q; let us denote them F' and G', so that we have the equality $(\mathrm{F}\,\mathrm{G})' = \mathrm{F}'\,\mathrm{G}'$. We claim that they have only ordinary multiple points and no tangents in common at points of intersection. Assume for contradiction the contrary. First assume that F' has a tangent ℓ with multiplicity $e > 1$ at some point. About here we may homogeneize (w.l.o.g. in Z) and so write the two polynomials as the sum of forms,

$$\mathrm{F}'[X, Y, 1] = \ell^k \prod_i \ell_i^{e_i} + \cdots, \qquad \mathrm{G}'[X, Y, 1] = \mathrm{G}_m + \cdots$$

And so,

$$(\mathrm{F}\,\mathrm{G})'[X, Y, 1] = \mathrm{F}'[X, Y, 1]\, \mathrm{G}'[X, Y, 1] = \mathrm{G}_m\, \ell^k \prod_i \ell_i^{e_i} + \cdots$$

And thereby $(\mathrm{F}\,\mathrm{G})'$ also has a singularity at the point, contradicting it being resolved. Now assume that F' and G' share a tangent ℓ in common at a point of intersection. About here we may homogeneize again and write,

$$\mathrm{F}'[X, Y, 1] = \ell \prod_i \mathrm{f}_i + \cdots, \qquad \mathrm{G}'[X, Y, 1] = \ell \prod_j \mathrm{g}_j + \cdots$$

And so,

$$(\mathrm{F}\,\mathrm{G})'[X,Y,1] = \mathrm{F}'[X,Y,1]\,\mathrm{G}'[X,Y,1] = \ell^2 \prod_i \mathrm{f}_i \prod_j \mathrm{g}_j + \cdots$$

And thereby $(\mathrm{F}\,\mathrm{G})'$ has a singularity at the point, contradicting it being resolved.

We prove the latter part of the proposition by induction on s. The base case $s = 1$ or $\mathrm{F} = \mathrm{F}'$, $\mathrm{G} = \mathrm{G}'$ is simply Bezout's Theorem. Using the inductive hypothesis we have,

$$\deg \mathrm{F}_2 \deg \mathrm{G}_2 = \sum_{\mathbf{q}} \mathrm{m}_{\mathbf{q}}[\mathrm{F}_2]\,\mathrm{m}_{\mathbf{q}}[\mathrm{G}_2] + \sum_{i>2}^{k} \sum_{j=1}^{s_i} \mathrm{m}_{\mathbf{q}_{i,j}}[\mathrm{F}_i]\,\mathrm{m}_{\mathbf{q}_{i,j}}[\mathrm{G}_i]$$

On the l.h.s., w.l.o.g. using a projective transformation so that $\mathbf{p}$ is the only fundamental point on $\{\mathrm{F} = 0\}$ and $\{\mathrm{G} = 0\}$, we have the equality,

$$\begin{aligned}\deg \mathrm{F}_2 \deg \mathrm{G}_2 &= (2\deg \mathrm{F} - \mathrm{m}_{\mathbf{p}}[\mathrm{F}])(2\deg \mathrm{G} - \mathrm{m}_{\mathbf{p}}[\mathrm{G}]) \\ &= 4\deg \mathrm{F} \deg \mathrm{G} - 2\deg \mathrm{F}\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}] - 2\deg \mathrm{G}\,\mathrm{m}_{\mathbf{p}}[\mathrm{F}] + \mathrm{m}_{\mathbf{p}}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}]\end{aligned}$$

Meanwhile we may categorize and expand,

$$\begin{aligned}\sum_{\mathbf{q}} \mathrm{m}_{\mathbf{q}}[\mathrm{F}_2]\,\mathrm{m}_{\mathbf{q}}[\mathrm{G}_2] &= \mathrm{m}_{\mathbf{p}}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}] + \mathrm{m}_{\mathbf{p}'}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}'}[\mathrm{G}] + \mathrm{m}_{\mathbf{p}''}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}''}[\mathrm{G}] \\ &\quad + \sum_{j=1}^{s_2} \mathrm{m}_{\mathbf{q}_{2,j}}[\mathrm{F}_2]\,\mathrm{m}_{\mathbf{q}_{2,j}}[\mathrm{G}_2]\end{aligned}$$

Namely the first term may be expanded,

$$\begin{aligned}\sum_{\alpha} \mathrm{m}_{\alpha}[\mathrm{F}]\,\mathrm{m}_{\alpha}[\mathrm{G}] &= \deg \mathrm{F} \deg \mathrm{G} + 2(\deg \mathrm{F} - \mathrm{m}_{\mathbf{p}}[\mathrm{F}])(\deg \mathrm{G} - \mathrm{m}_{\mathbf{p}}[\mathrm{G}]) \\ &= 3\deg \mathrm{F} \deg \mathrm{G} - 2\deg \mathrm{F}\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}] - 2\deg \mathrm{G}\,\mathrm{m}_{\mathbf{p}}[\mathrm{F}] + 2\,\mathrm{m}_{\mathbf{p}}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}]\end{aligned}$$

So through rearranging,

$$\deg \mathrm{F} \deg \mathrm{G} = \mathrm{m}_{\mathbf{p}}[\mathrm{F}]\,\mathrm{m}_{\mathbf{p}}[\mathrm{G}] + \sum_{i>1}^{k} \sum_{j=1}^{s_i} \mathrm{m}_{\mathbf{q}_{i,j}}[\mathrm{F}_i]\,\mathrm{m}_{\mathbf{q}_{i,j}}[\mathrm{G}_i]$$

As desired.

■